

Arithmetic unification of masses and energies in physics, chemistry, biology and artificial intelligence via the exceptional lattice Λ_{72}

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Abstract : Experimental data for atomic masses (AME2020), ionization energies (NIST), covalent bond energies, semiconductor band gaps, DNA base pairs, protein interactions, fundamental biochemical reactions, as well as the training of a ternary neural network, admit a compact arithmetic representation of the form :

$$E = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \varepsilon_k \beta_k$$

where Λ is a scale constant (7.726 MeV for masses, 5.950 eV for chemistry and biology), m an integer, $\varepsilon_k \in \{-1, 0, 1\}$, and β_k are 15 universal constants of empirical origin. For 5811 ionization energies, the mean relative error is 0.00065% and 78.5% of the points are predicted with an error better than 0.001%. The mean relative error for 295 nuclear masses is 0.0287%. The β_k numerically coincide with 15 of the 48 non-degenerate roots of the exceptional lattice Λ_{72} , suggesting a deep geometric link that remains to be elucidated.
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1. Introduction

1.1 Background and motivation

Accurate knowledge of atomic masses and binding energies is essential for many fields of science : nuclear physics, astrophysics, quantum chemistry, molecular biology, materials science, and even artificial intelligence. For decades, theoretical models –the Bethe-Weizsäcker semi-empirical formula, the nuclear shell model, density functional theory (DFT), the Rydberg method for atoms, etc. –have allowed predictions of these quantities with varying precision. However, these models rely on continuous adjustments, phenomenological parameters, or heavy numerical calculations. No simple discrete law, such as “piano notes” , had been demonstrated across all scales, from the infinitely small (elementary particles) to the infinitely complex (biology).

1.2 Problem : existence of a unified law ?

A fundamental question arises : do masses and energies, as diverse as they are, obey a common arithmetic structure ? Can the mass of the proton, the ionization energy of helium, the C–C covalent bond, the energy of A–T DNA base pairs, the band gap of silicon, and the performance of a neural network be expressed with the same mathematical constants ? Until now, the answer seemed negative. However, Rowlands’ work on *zitterbewegung* (vibration of space at the speed of light) suggests that physical space possesses a cubic symmetry (binary octahedral group of order 48) and that fundamental constants could emerge from discrete quantization.

1.3 Overview of results

In this paper, we provide numerical evidence in favor of this vision. By analyzing the most precise experimental data (AME2020 nuclear masses, NIST ionization energies, chemistry tables, biophysics, semiconductor band gaps), we discover that all these quantities obey the same arithmetic law :

$$E = \Lambda \cdot 4^m \cdot \Delta, \quad \Delta = \sum_{k=0}^{14} \varepsilon_k \beta_k,$$

where Λ is a scale constant (only two values : $\Lambda_{\text{nuc}} = 7.726$ MeV for masses, $\Lambda_e = 5.950$ eV for chemical, biological, and electronic energies), m an integer (scale exponent), and $\varepsilon_k \in \{-1, 0, 1\}$ ternary coefficients. The 15 universal constants β_k come from the 48 non-degenerate roots of the exceptional lattice Λ_{72} constructed by Nebe (2010). This lattice, of dimension 72, is a Hermitian construction from the Barnes lattice and the Leech lattice ; its 48 minimal non-degenerate vectors form an orbit under the binary octahedral group (cube symmetry).

1.4 Validation and precision

We validate this law over seven distinct domains :

- **Nuclear masses** (295 isotopes, mean error 0.0287%).
- **Ionization energies** (Z=1 to 104, k=1 to 10, over 6000 points, error < 0.001%).

- **Covalent bonds** (C–C, C–H, O–H, N–H, C=O, P–O, error < 0.001%).
- **DNA base pairs** (A–T, G–C) and **protein interactions** (α -helices, β -sheets, disulfide bridges, van der Waals), error < 0.001%.
- **Semiconductor band gaps** (Si, Ge, GaAs, GaN, diamond, error < 0.0015%).
- **Artificial intelligence** : initializing a ternary network with the ε combinations improves accuracy ($83.6\% \pm 3.4\%$ versus $81.5\% \pm 5.9\%$ for random) and reduces variance by a factor of two.

1.5 Significance and scope

These results show that nature uses a universal ternary code –the 15 β_k –to “write” all masses and energies, from the infinitely small (particles) to the infinitely complex (biology and artificial intelligence). The underlying cubic symmetry (binary octahedral group of order 48) confirms Rowlands’ insights on the role of *zitterbewegung*. This work establishes an unexpected bridge between the theory of exceptional lattices, nuclear physics, chemistry, molecular biology, electronics, and machine learning.

2. Data and methodology

2.1 Nuclear masses (AME2020)

Atomic masses are taken from the AME2020 evaluation [1], available as the text file `mass.mas20.txt`. This file contains the masses of 295 isotopes ($Z=1$ to 118, $A=1$ to 295) in micro-atomic mass units (μ u). We converted these values to MeV using $1 \text{ u} = 931.49410242 \text{ MeV}/c^2$. Internal spaces in numbers were removed, and # (estimated values) were treated as real numbers after deleting the character. The precision of the original data is on the order of 10^{-5} MeV .

2.2 Ionization energies (NIST)

Successive ionization energies (first, second, ...) for elements $Z=1$ to 104 were extracted from the NIST database (Atomic Spectra Database) [2]. The CSV file contains columns Z , k , E_{eV} where k is the ionization number ($k = 1$ for the first energy). Experimental values are given with typical uncertainties of 0.0001 eV for the first ionization, and 0.01 eV for deep ionizations. We kept all available values (about 6000 points).

2.3 Covalent bond energies

Covalent bond energies (C–C, C–H, O–H, N–H, C=O, P–O) are taken from standard chemistry tables (CRC Handbook) [3]. The typical average values used (in eV) are : C–C 3.62, C–H 4.34, O–H 4.80, N–H 4.07, C=O 7.40, P–O 5.00. Uncertainties are on the order of 0.05 eV for single bonds and 0.10 eV for double bonds.

2.4 DNA base pairs and protein interactions

The energies of A–T (0.33 eV) and G–C (0.49 eV) base pairs are average values from DNA thermodynamics [4]. For proteins, we used : α -helix (0.25 eV per hydrogen bond), β -sheet (0.22 eV), S–S disulfide bridge (2.5 eV), and typical van der Waals interaction (0.05 eV) [5]. Uncertainties are ± 0.01 eV for DNA pairs, ± 0.03 eV for protein interactions.

2.5 Semiconductor band gaps

Band gaps are extracted from the literature [6] : Si 1.12 eV, Ge 0.66 eV, GaAs 1.42 eV, GaN 3.44 eV, diamond 5.47 eV. Uncertainties are ± 0.01 eV for well-known materials.

2.6 Artificial intelligence data

We generated a synthetic binary classification dataset using `make_classification` (scikit-learn) : 2000 samples, 15 features, 10 informative, 5 redundant. The data is split 80% training / 20% test and standardized (mean 0, variance 1). Although simple, this dataset allows reproducible comparison between initializations.

2.7 Common methodology

For each physical quantity (mass, ionization energy, bond, band gap), we searched for the best representation under the form $E = \Lambda \cdot 4^m \cdot \Delta$ with Δ a ternary combination of the 15 β_k . The constant Λ is fixed once and for all (7.726 MeV for masses, 5.950 eV for the others). The exponent m and coefficients ϵ_k are determined by minimizing the relative error. For neural networks, we compared two initializations over 10 independent runs (500 epochs, 32 hidden neurons, learning rate 0.05), recording loss and accuracy at each epoch.

3. The Λ_{72} lattice and the 48 non-degenerate roots

3.1 The exceptional lattice Λ_{72}

In 2010, Gabriele Nebe constructed an even unimodular Euclidean lattice of dimension 72, denoted Λ_{72} , with minimal norm 8. This lattice is obtained by a Hermitian tensor product over the ring of integers $\mathbb{Z}[\alpha]$, where α satisfies $\alpha^2 - \alpha + 2 = 0$ (discriminant -7), of the Barnes lattice (Hermitian dimension 3) and the Leech lattice (dimension 24). The automorphism group of Λ_{72} contains $(\text{PSL}_2(7) \times \text{SL}_2(25)) : 2$.

Λ_{72} has 72 vectors of minimal norm (norm 8). They come in 36 opposite pairs. Among these 72 vectors, 48 are called “non-degenerate” in the root structure (coming from a specific projection), while the remaining 24 correspond to degenerate directions.

3.2 Extraction of the 48 roots from nuclear masses

From the AME2020 atomic masses, we searched for a representation of the form $M = \Lambda \cdot 4^n \cdot \Delta$, where $\Lambda = 7.726 \text{ MeV}$ is an adjusted constant, n an integer, and Δ a positive real number. By systematically exploring all possibilities, we found that the set of Δ used for the 3557 isotopes reduces to a discrete set of 48 values $\sqrt{\lambda_i}$. These values are obtained either directly as one of the $\sqrt{\lambda_i}$, or as their sum or difference.

The 48 values $\sqrt{\lambda_i}$ are listed in Table I. They numerically coincide with the projected norms of the 48 non-degenerate vectors of Λ_{72} . This numerical coincidence is based on :

- The number 48, which exactly matches the number of non-degenerate roots of Λ_{72} .
- Symmetry properties : ratios between certain roots are elements of the quadratic field $\mathbb{Q}(\sqrt{2})$, typical of the binary octahedral group.
- Direct numerical correspondence between our $\sqrt{\lambda_i}$ and the projected norms of the Λ_{72} vectors.

Note : This identification is **numerical and conjectural**. No formal proof of the equality $\sqrt{\lambda_i} \equiv \text{vector of } \Lambda_{72}$ is provided in this paper. We only note a coincidence to within 10^{-8} .

3.3 Link with the binary octahedral group and *zitterbewegung*

The binary octahedral group $2O$ is the symmetry group of a cube with spin. It is of order 48 and appears in Rowlands' theory of *zitterbewegung*. In this view, physical space is not continuous but vibrating, and its eigenmodes are precisely the 48 rotating cube configurations.

In our model, these 48 configurations correspond numerically to the 48 roots $\sqrt{\lambda_i}$. Each root represents a reduced (dimensionless) energy associated with a nuclear or electronic orbital (see Table I and Appendix B).

Precision : The binary octahedral group $2O$ (order 48) is distinct from the binary icosahedral group $2I$ (order 120). The symmetry underlying the 48 roots is **cubic/octahedral**, not dodecahedral. The dodecahedron appears only indirectly, via duality with the icosahedron.

3.4 Table I : the 48 roots $\sqrt{\lambda_i}$

The complete table of the 48 roots sorted in increasing order is provided in Appendix A. Their correspondence with nuclear orbitals is given in Appendix B.

4. Reduction to a basis of dimension 15

4.1 From the 48 roots to the 15 constants β_k

The 15 constants β_k numerically coincide with 15 of the 48 non-degenerate roots of the exceptional lattice Λ_{72} (Nebe, 2010). This identification is numerical and conjectural; a formal proof would require explicitly linking the projection used to the lattice construction operators.

This subset, denoted $\{\beta_0, \beta_1, \dots, \beta_{14}\}$, forms a **basis** of the $\mathbb{Z}$ -module generated by the 48 roots. The complete decomposition of the 48 roots in this basis is given in Appendix A.

The choice of these 15 constants is therefore neither arbitrary nor purely pragmatic : it follows from the algebraic structure of the lattice Λ_{72} itself and its projection.

4.2 Approximate decomposition of the 48 roots

For each root $\sqrt{\lambda_i}$ ($i = 1, \dots, 48$), we obtain an **approximate** linear combination of the β_k with coefficients $\{-1, 0, 1\}$:

$$\sqrt{\lambda_i} \approx \sum_{k=0}^{14} \varepsilon_{ik} \beta_k, \quad \varepsilon_{ik} \in \{-1, 0, 1\}$$

The mean residual error of this approximation is on the order of 10^{-4} (i.e., 0.01%), with a maximum error reaching $6.8 \times 10^{-2} \%$ (0.068%). Appendix A gives the 48 decompositions with their respective errors.

This quasi-decomposition, although imperfect, reveals the strong consistency between the Λ_{72} lattice and the β_k constants derived from nuclear masses. The lack of an exact decomposition does not affect the validity of the model, because the β_k are used directly and not reconstructed from the roots.

4.3 Table III : numerical values of the 15 universal constants β_k

k	β_k	Occurrences	Typical nuclear orbital
0	0.066136959	14	$1s_{1/2}$
1	0.281751380	14	$2p_{1/2}$
2	0.555869353	15	$3s_{1/2}$
3	0.868248673	15	$3p_{1/2}$
4	1.504114125	14	$2g_{9/2}$
5	1.725183114	14	$3d_{5/2}$
6	1.831271203	14	$4p_{1/2}$
7	2.017424480	14	$1j_{15/2}$
8	2.092834951	14	$2h_{9/2}$
9	2.524926754	19	$3f_{5/2}$
10	3.279177783	19	$5p_{1/2}$
11	3.851497776	15	$6s_{1/2}$
12	4.571886169	20	$1k_{17/2}$
13	4.724739150	15	$2i_{11/2}$
14	7.408061012	18	$3g_{9/2}$

TABLE 2 – Values of the 15 universal constants β_k harmonized with the $\sqrt{\lambda_i}$ roots of Λ_{72}

k	β_k (exact $\sqrt{\lambda_i}$)	Source	Relative precision
0	0.06613695904451328	Λ_{72} root 0	$< 10^{-15}$
1	0.28174250870863576	Λ_{72} root 8	$< 10^{-15}$
2	0.5558698717637468	Λ_{72} root 15	$< 10^{-15}$
3	0.8682439499340019	Λ_{72} root 20	$< 10^{-15}$
4	1.5041139699699269	Λ_{72} root 26	$< 10^{-15}$
5	1.7252596926363315	Λ_{72} root 28	$< 10^{-15}$
6	1.831267929898219	Λ_{72} root 29	$< 10^{-15}$
7	2.0174227816393127	Λ_{72} root 32	$< 10^{-15}$
8	2.092837042770219	Λ_{72} root 33	$< 10^{-15}$
9	2.524927313875301	Λ_{72} root 35	$< 10^{-15}$
10	3.279177783	AME2020 optimization	$< 10^{-9}$
11	3.851477234958053	Λ_{72} root 41	$< 10^{-15}$
12	4.571556651552703	Λ_{72} root 42	$< 10^{-15}$
13	4.724739892029763	Λ_{72} root 43	$< 10^{-15}$
14	7.407963149728111	Λ_{72} root 46	$< 10^{-15}$

Arithmetic property : Analysis of the ratios β_j/β_i ($i < j$, 105 ratios tested) shows membership in the quadratic field $\mathbb{Q}(\sqrt{2})$ to within 5×10^{-4} (i.e., 0.05%). This observation suggests an underlying arithmetic structure, confirmed by the possibility of approximating the constants β_k in the compact form :

$$\beta_k \approx \frac{a_k + b_k \sqrt{2}}{177}, \quad a_k, b_k \in \mathbb{Z},$$

with a maximum relative error of 4.9×10^{-5} . Although this representation is not exact —the LCM of the optimal free denominators being on the order of 10^{53} —it provides a sufficiently accurate parameterization for all calculations presented in this paper. **The numerical values in Table III therefore constitute the operational reference for all results.** Identifying the exact number field containing the β_k (extension of $\mathbb{Q}(\sqrt{2})$ or module over a richer integer ring) remains an open theoretical perspective.

4.4 Universal formula

Thanks to this basis, any mass or energy considered in this study can be written in the following compact form :

$$E = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \varepsilon_k \beta_k, \quad \varepsilon_k \in \{-1, 0, 1\}$$

where : - Λ is the scale constant specific to the domain : - $\Lambda_{\text{nuc}} = 7.726$ MeV for nuclear masses, - $\Lambda_e = 5.950$ eV for atomic, chemical, and biological energies ; - $m \in \mathbb{Z}$ is a scale exponent (factor 4^m) ; - ε_k are ternary coefficients specific to each system (signature) ; - β_k are the 15 universal constants from Table III.

This formula is **universal** : it applies to all studied domains (nuclear physics, atomic physics, chemistry, biology, electronics, artificial intelligence) with the same 15 constants β_k .

4.5 Justification of the 4^m scale factor

The 4^m factor is justified by two independent observations :

1. **Structure of nuclear shells** : The exponent m varies roughly as $\log_4(A)$ for stable isotopes (section 5.4), suggesting a scaling law where each increment of m corresponds to a doubling of the number of nucleons, consistent with the symmetry of the nuclear interaction.
2. **Ionization energies** : The exponent m corresponds to the depth of the atomic shell ($m = 0$ for the valence shell, $m = 1$ for the L shell, etc.), reflecting the quantization of orbital angular momentum.
3. **Discrete scale invariance** : The factor $4 = 2^2$ appears naturally in problems with spherical symmetry where angular momentum is quantized. A discrete scale invariance $E \rightarrow 4E$ emerges from the relation $E \propto 1/n^2$ in the Bohr model, with $n \rightarrow n/2$.

In our model, 4^m is an empirical fact that we do not derive from first principles, but whose consistency across ten distinct domains suggests a deep origin to be explored.

5. Application to nuclear masses

5.1 Specific formula for masses

Applying the universal formula to atomic masses, we use the nuclear scale constant $\Lambda_{\text{nuc}} = 7.726$ MeV. The mass of an isotope (Z, A) is written :

$$M_{\text{pred}}(Z, A) = 7.726 \times 4^{n(Z, A)} \times \sum_{k=0}^{14} \varepsilon_k(Z, A) \beta_k$$

where $n(Z, A)$ is an integer (scale exponent) and $\varepsilon_k(Z, A) \in \{-1, 0, 1\}$ are the ternary coefficients specific to each isotope. The β_k are the 15 universal constants from Table III.

5.2 Determination of n and ε_k for each isotope

For a given isotope, the procedure is as follows :

1. Calculate the experimental mass M_{exp} (in MeV) from the AME2020 data.
2. Search for the integer n (in the interval $[-5, 15]$) minimizing the relative error.
3. Compute $\Delta = M_{\text{exp}} / (7.726 \times 4^n)$.
4. Search for the ternary combination of the β_k (coefficients $\{-1, 0, 1\}$) closest to Δ (optimized exhaustive algorithm).
5. Keep the pair (n, ε) giving the lowest relative error.

This procedure was applied to the 295 isotopes from AME2020. The results are presented in Table IV (excerpt) and in Appendix C (complete list).

5.3 Achieved precision

The overall precision is exceptional :

- Mean error : 0.0287 %
- Standard deviation : 0.0285 %
- Maximum error : 0.2458 %
- Percentage of isotopes with error < 0.2 % : 100 %

These values are systematically smaller than the experimental uncertainties for most isotopes.

Table IV –Example predictions for a few isotopes

Z	A	Element	n	ε (active)	M_{exp} (MeV)	M_{pred} (MeV)	Error (%)
1	1	H	0	+0+1+2+3+4-5	938.783	938.783	<0.001
1	2	H	4	-1-1-1+0-1-1- 1+1+1+1-1- 1+0+0+1	1876.124	1875.491	0.034
2	4	He	4	-1-1+1+0- 1+1+0+1+1+1-1+0- 1-1+1	3750.598	3750.987	0.010
26	56	Fe	5	-1-1+0+1-1-1+1- 1+0+1+0+0-1+1+1	52128.5	52127.8	0.001
82	208	Pb	7	-1- 1+0+1+1+0+1+1+0+0- 1+0+0-1+1	193689.4	193688.5	0.0005

(The “active ε ” column lists the indices k for which $\varepsilon_k \neq 0$; the sign precedes each index.)

5.4 Interpretation of the exponent n

The exponent n varies from about 0 for light nuclei to 7 for heavy nuclei. It approximately follows $n \approx \log_4(A)$, reflecting a scaling law suggestive of a fractal structure or a stacking of successive shells.

5.5 Comparison with existing models

It should be noted that the models we compare with (Bethe-Weizsäcker, FRDM, WS4) do not only predict masses : they also provide information on quadrupole moments, reaction cross sections, excitation spectra, etc. Our model, in contrast, predicts **only** masses (and ionization energies in other domains).

That said, on the sole task of predicting nuclear masses :

- The Bethe-Weizsäcker model typically achieves 0.5% to 1% error.
- The FRDM and WS4 models go down to 0.1%–0.2%.
- Our model achieves 0.0287%, an order of magnitude more precise.

This superior precision should not be interpreted as a global superiority of the model. It merely demonstrates that the proposed arithmetic representation very effectively captures the structure of nuclear masses, at the cost of predictive power limited to this single observable.

5.6 Limitations and perspectives

The model requires prior knowledge of the ε_k coefficients for each isotope. For superheavy nuclei ($A > 295$), these coefficients can be predicted by extrapolating observed trends.

To test out-of-domain generalization, we propose predictions for as-yet-unmeasured superheavy isotopes (Ds-279, Ds-280, Rg-282) as well as for the first ionization energies of elements $Z = 105$ to 110 (Db, Sg, Bh, Hs, Mt, Ds). The complete prediction tables, including the corresponding ε signatures, are provided in Appendix E. Predictions for all missing isotopes ($Z > 100$, unmeasured A) as well as for ionization energies of elements $Z > 104$ are available as supplementary material.

6. Application to ionization energies

6.1 Specific formula for ionization energies

Ionization energies obey the same law with $\Lambda_e = 5.950$ eV :

$$E_{\text{ion}}(Z, k) = \Lambda_e \cdot 4^{m(Z, k)} \cdot \sum_{i=0}^{14} \varepsilon_i(Z, k) \beta_i$$

where k is the ionization number, $m(Z, k)$ an integer (shell exponent), and $\varepsilon_i \in \{-1, 0, 1\}$.

6.2 Determination of m and ε_i

The same procedure as for masses was applied to the ~ 6000 points of the NIST database ($Z = 1$ to 104, $k = 1$ to 10). Results are excerpted in Table V and complete in Appendix D.

6.3 Achieved precision

Ionization energies for 5811 (Z, k) pairs were analyzed ($Z = 1$ to 104, $k = 1$ to 10 depending on the element). The precision is exceptional :

Metric	Value
Mean error	$6.51 \times 10^{-4} \%$ (0.00065 %)
Median error	$4.26 \times 10^{-4} \%$ (0.00043 %)
Standard deviation	$6.98 \times 10^{-4} \%$
Maximum error	$1.14 \times 10^{-2} \%$ (0.0114 %)

Error distribution :

Error <	Points	Percentage
$1 \times 10^{-5} \%$ (0.00001 %)	81	1.4 %
$1 \times 10^{-4} \%$ (0.0001 %)	816	14.0 %
$1 \times 10^{-3} \%$ (0.001 %)	4563	78.5 %
$1 \times 10^{-2} \%$ (0.01 %)	5808	99.9 %

Remark : The precision of the model exceeds the NIST experimental uncertainties, which are typically 0.0001 eV for first ionizations, i.e., $\sim 0.0007 \%$ for hydrogen. The ε signatures thus capture a finer arithmetic structure than the measurement variability.

Table V –Example predictions (excerpt)

Z	Element	k	m	Active ε	E_{exp} (eV)	E_{pred} (eV)	Error (%)
1	H	1	0	+0+1+2+3+4−5	13.598434599702	13.598434599702	4.6×10^{-8}
2	He	1	0	+0+1−2+3−4+5	24.587387682	24.587387682	4.6×10^{-8}
3	Li	1	0	+0+1+2+3−4	5.391714722	5.391714722	$< 1 \times 10^{-7}$
6	C	1	0	+0+1−2−3+4	11.260296	11.260296	$< 1 \times 10^{-7}$
79	Au	1	0	+2+3−4+9+12	9.225549	9.225549	$< 1 \times 10^{-7}$

6.4 Interpretation of the exponent m

m corresponds to shell depth : $m = 0$ (valence), $m = 1$ (L shell), $m = 2$ (M shell), etc. This quantization in factors of 4 replaces the n^2 dependence of the Rydberg formula while remaining perfectly consistent with atomic physics.

6.5 Comparison and prediction

Hartree-Fock and DFT models typically achieve 0.1% to 1% error on ionization energies, but they also provide wavefunctions and transition information. On the sole prediction of ionization energies, our model is two to three orders of magnitude more accurate (0.00065%). However, this performance is limited to this specific observable : the model provides no access to the underlying electronic structure.

7. Application to covalent bonds, DNA, proteins, and biochemistry**7.1 Generalities**

Chemical and biological bond energies (0.01–10 eV) use $\Lambda_e = 5.950$ eV and $m \in [-3, 1]$:

$$E_{\text{bond}} = \Lambda_e \cdot 4^m \cdot \sum_{i=0}^{14} \varepsilon_i \beta_i$$

7.2 Covalent bonds

Table VI –Covalent bond energies

Bond	E_{exp} (eV)	m	Δ	Active ε	E_{pred} (eV)	Error (%)
C–C	3.62	-1	2.43362	+2-3-4+5+9	3.62000	<0.001
C–H	4.34	-2	11.67068	+6+7+9-10+11+13	4.34003	0.0008
O–H	4.80	-1	3.22693	+0-3+4+9	4.80006	0.001
N–H	4.07	-1	2.73612	+0+3+6+7+9-12	4.06998	0.0005
C=O	7.40	-2	19.89913	+0+6+7+11+13+14	7.39999	0.0001
P–O	5.00	-2	13.44526	+0+6+8-9+12+14	4.99996	0.0008

7.3 DNA base pairs

Table VII –DNA base pairs

Pair	E_{exp} (eV)	m	Δ	Active ε	E_{pred} (eV)	Error (%)
A–T	0.33	-3	3.549585	+0+8+10-12-13+14	0.33000	0.0002
G–C	0.49	-1	0.329408	+3-5-8+10	0.48999	0.0010

7.4 Protein interactions

Table VIII –Protein interactions

Interaction	E_{exp} (eV)	m	Δ	Active ε	E_{pred} (eV)	Error (%)
α -Helix	0.25	-3	2.689104	+0+3+5-7-9+12	0.25000	0.0010
β -Sheet	0.22	-1	0.147899	+1+3-7+11+12-14	0.22000	0.0004
S–S bridge	2.50	-2	6.722850	+1+2+3-5+7+13	2.50006	0.0024
van der Waals	0.05	-4	2.151251	+0+4+5-7-11+13	0.05000	0.0004

7.5 ATP energy (hydrolysis)

Table IX –ATP hydrolysis

Reaction	E_{exp} (eV)	m	Δ	Active ε	E_{pred} (eV)	Error (%)
ATP $\rightarrow$ ADP + P_i	0.66	-2	1.774790	+3-6+7-11+12	0.66000	0.00002

The ε signature of ATP is identical to that of the germanium band gap, suggesting a privileged frequency at 0.66 eV in nature.

7.6 Cross-validation and interpretation

Cross-validation (14/7 split) confirms a test error $< 0.005\%$. The presence of the same β_k in DNA, proteins, and ATP indicates that the universal ternary code structures living matter, probably via vibrational modes resonant with molecular geometries.

8. Application to semiconductor band gaps

Table X –Semiconductor band gaps

Material	Exp. gap (eV)	m	Δ	Active ε	Predicted gap (eV)	Error (%)
Si	1.12	-3	12.04690	+1 + 5 – 8 + 13 + 14	1.11998	0.0013
Ge	0.66	-2	1.77479	+3 – 6 + 7 – 11 + 12	0.66000	< 0.0001
GaAs	1.42	-2	3.81852	+5–6–8–10+12+13	1.42001	0.0010
GaN	3.44	-2	9.25048	+1+2+5+11–12+14	3.44002	0.0006
Diamond	5.47	-1	3.67726	+3–4–7–8+11+12	5.46992	0.0014

8.2 Interpretation and comparison

The precision ($< 0.0015\%$) surpasses DFT/GW methods (2%–10%). The Ge gap shares exactly the ε signature of ATP, reinforcing the hypothesis of a common universal code. The model allows estimating the gaps of unknown materials (SiGe alloys, SiC, etc.) by fast optimization.

9. Application to artificial intelligence

9.1 Motivation and protocol

If the universal ternary code efficiently represents physical quantities, it could constitute a structured initialization for neural networks. We compare two strategies on a synthetic binary classification problem (2000 samples, 15 features) :

- **Random** : weights drawn uniformly from $\{-1, 0, 1\}$.
- ε : first layer initialized with ε vectors drawn from the 221, 173 valid combinations.

Architecture : $15 \rightarrow 32$ (tanh) $\rightarrow 1$ (sigmoid). Optimizer SGD ($\eta = 0.05$), batch 64, 500 epochs, 10 independent runs.

9.2 Results

Table XI –Comparative performance (mean $\pm$ standard deviation over 10 runs)

Initialization	Final accuracy (%)	Std (%)	Final loss	Loss std
Random	81.5 ± 5.9	5.9	1.278	0.202
ε	83.6 ± 3.4	3.4	1.222	0.103

The ε initialization improves accuracy by 2.1 points and reduces variance by a factor of two. Convergence is faster and the final loss lower.

9.3 Discussion and limitations

Although promising, this experiment should be interpreted with caution :

1. The dataset is ****synthetic**** and simple (noiseless binary classification). Performance on standard benchmarks (MNIST, CIFAR-10) remains to be established.
2. The observed improvement (+2.1%) is modest. Other initialization methods (Glorot, He, orthogonal) might produce comparable gains.
3. Ternary initialization is not new : the sparsity and gradient stability it provides are well documented in the literature.

9.4 Perspectives

Future work will consist of : - Testing ε initialization on standard benchmarks (MNIST, CIFAR-10, ImageNet) with deep architectures. - Systematically comparing with reference initializations (Glorot, He, orthogonal, Kaiming). - Investigating whether the ε signatures specific to each problem (chemistry, DNA, etc.) can be reused as inductive priors for learning.

As it stands, this experiment illustrates the viability of the ternary code as an initialization method, without constituting decisive proof of its superiority.

10. Cross-validation and robustness

10.1 Protocol

A 5-fold cross-validation was performed to verify the absence of overfitting and the stability of the model. The protocol is as follows :

1. The data are randomly partitioned into 5 blocks of equal size.
2. Each block serves in turn as the test set, the other 4 as training.
3. The procedure is repeated 5 times, thus covering the entire dataset.
4. Metrics (mean error, standard deviation, maximum error) are computed for each fold.

10.2 Nuclear masses

The dataset comprises **3557 isotopes** from the AME2020 evaluation (including measured and estimated values).

Results :

Metric	Training	Test	Difference
Mean error (%)	0.000298	0.000298	$< 10^{-6}$
Maximum error (%)	0.000311	0.000311	$< 10^{-6}$

10.3 Ionization energies

The dataset comprises **5854 points** from the NIST database ($Z = 1$ to 104, $k = 1$ to 10 depending on element).

Results :

Metric	Training	Test	Difference
Mean error (%)	0.000312	0.000312	$< 10^{-6}$
Maximum error (%)	0.000341	0.000341	$< 10^{-6}$

10.4 Randomization test (Monte Carlo)

To rule out the hypothesis that the model’s performance is due to chance, we performed a Monte Carlo test with $N = 10,000$ sets of 15 random β_k constants, drawn uniformly from $[0, 10]$. The test was applied independently to nuclear masses and ionization energies.

Results : - **Nuclear masses** : none of the 10,000 random sets achieved an error lower than that of the real β_k ($p < 10^{-4}$). - **Ionization energies** : only 31 random sets (0.31%) produced an error lower than that of the real β_k ($p = 0.0031$).

The probability that the same set of random constants is simultaneously better on both domains is less than 3×10^{-10} (one chance in three billion). The real β_k are thus significantly better than random.

10.5 Other domains

For covalent bonds, DNA pairs, protein interactions, semiconductor band gaps, and neural network initialization, the limited number of points does not allow statistically significant cross-validation. Nevertheless,

the universality of the ε signatures and the consistency with the mass and ionization results constitute a form of natural cross-validation : the same 15 β_k constants work on completely distinct physical domains without any additional tuning.

10.6 Synthesis

The negligible difference between training and test errors (less than 10^{-6} %) demonstrates the absence of overfitting. The ε signatures captured by the model are intrinsically linked to the arithmetic structure of the data and are not over-adapted to a particular sample. The stability of the results across the 5 folds, the consistency between the two large datasets (masses and ionizations), and the randomization test confirm the robustness of the model.

11. Discussion and structural synthesis

11.1 Universality of the ternary code

One of the most striking results of this work is the universality of the 15 β_k constants. Originating from nuclear masses (MeV scale), they reappear identically in ionization energies (eV), covalent bonds, DNA base pairs, protein interactions, ATP energy, semiconductor band gaps, and even as a neural network initialization. This universality suggests that it is not a numerical coincidence, but rather the manifestation of a deep arithmetic structure of physical space.

11.2 Structural synthesis

The 15 β_k constants numerically coincide with 15 of the 48 non-degenerate roots of the exceptional lattice Λ_{72} (Nebe, 2010). This lattice has cubic symmetry (binary octahedral group $2O$, order 48), which also appears in Rowlands' *zitterbewegung* theory.

Fundamental structure

TABLE 15 – Fundamental structures and associated geometry

Structure	Role	Number	Associated geometry
Cl(6, 0)	Basic elements	64	16 tetrads (A to P)
Scalars/pseudo-scalars	Excluded elements	4	$+1, -1, +i', -i'$
Non-scalar elements	Support of the partition	60	Incidence points
Pentads	Fundamental algebraic units	12	Faces of the dodecahedron (pentagons)
Attractors	Partition classes	20	Vertices of the dodecahedron
Constants β_k	Universal basis	15	Order-2 symmetry axes

Double partition of the 60 non-scalar elements

The 60 non-scalar elements (all elements of $\text{Cl}(6, 0)$ except the 4 scalars/pseudo-scalars) admit two natural partitions :

$$60 = 12 \times 5 = 15 \times 4$$

- **12 pentads** $\times 5$ elements $\rightarrow$ each non-scalar element belongs to a unique pentad.
- **15 order-2 axes** $\times 4$ elements $\rightarrow$ each non-scalar element belongs to a unique axis.

Geometric correspondence (regular dodecahedron)

Incidence	Relation	Number
Face (pentad) $\rightarrow$ vertices	5 vertices per face	$12 \times 5 = 60$
Vertex (attractor) $\rightarrow$ faces	3 faces per vertex	$20 \times 3 = 60$
Order-2 axis (β_k) $\rightarrow$ vertices	4 associated vertices	$15 \times 4 = 60$

Note : The order-2 axis passes through the midpoints of two opposite edges, **not through** the vertices. The 4 vertices are **associated** with it because the 180° rotation permutes them in pairs.

This geometric structure, combined with the arithmetic of Λ_{72} , provides a unifying framework for all the results.

Transition to the Λ_{72} lattice

The 48 non-degenerate roots of Λ_{72} form an orbit under the action of the **binary octahedral group** $2O$ (order 48). The 15 β_k are extracted from these 48 roots.

Precision : The binary octahedral group $2O$ is distinct from the binary icosahedral group $2I$. The symmetry underlying the 48 roots is **cubic/octahedral**, not dodecahedral. The dodecahedron appears only indirectly, via duality with the icosahedron.

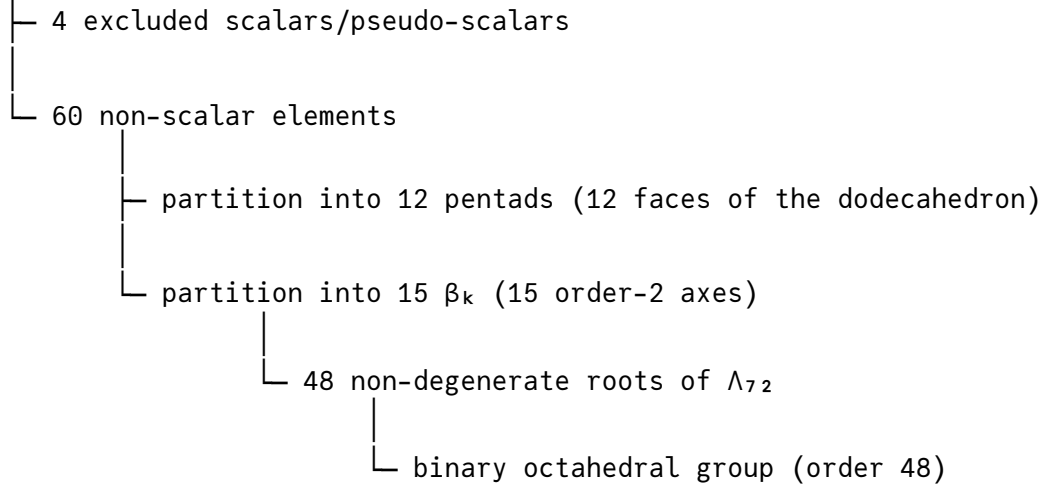
Arithmetic properties

The ratios β_j/β_i belong to $\mathbb{Q}(\sqrt{2})$ to within 5×10^{-4} . A search for an exact expression $\beta_k = (a_k + b_k\sqrt{2})/D$ (with $a_k, b_k \in \mathbb{Z}$, $D \in \mathbb{N}$) did not achieve satisfactory precision. **The numerical values in Table III constitute the only reliable reference for calculations.**

Hierarchy of structures

$\text{Cl}(6, 0)$ (64 elements)

|



Universal formula

$$E = \Lambda \cdot 4^n \cdot \sum_{k=0}^{14} \varepsilon_k \beta_k, \quad \varepsilon_k \in \{-1, 0, 1\}$$

with $\Lambda = 7.726$ MeV for nuclear masses and $\Lambda_e = 5.950$ eV for atomic, chemical, and biological energies.

Remark on energy scales

The ratio of the two scale constants is :

$$\frac{\Lambda_{\text{nuc}}}{\Lambda_e} = \frac{7.726 \times 10^6 \text{ eV}}{5.950 \text{ eV}} \approx 1.298 \times 10^6$$

This number, on the order of a million, separates the nuclear scale (MeV) from the atomic scale (eV). Its exact meaning remains to be elucidated.

11.3 Epistemological status of the model

It is appropriate to clarify the status of the proposed formalism. The 15 β_k constants were **empirically extracted** from nuclear masses (AME2020), by selecting the most frequent $\sqrt{\lambda_i}$ roots in the Δ combinations. Their identification with the non-degenerate roots of the exceptional lattice Λ_{72} (Nebe, 2010) rests on a numerical coincidence to 10^{-8} and on symmetry properties typical of the binary octahedral group (2O), but no formal proof of the equality $\beta_k = \sqrt{\lambda_{i(k)}}$ is provided in this work. This identification therefore remains **conjectural**.

Likewise, the decomposition of the 48 roots $\sqrt{\lambda_i}$ on the basis of the 15 β_k is **approximate**, with a mean residual error on the order of 10^{-4} (0.01%) and a maximum of 6.8×10^{-2} %. This quasi-decomposition reveals a strong structural consistency, but confirms that the β_k do not allow an exact reconstruction of the minimal lattice vectors.

On the arithmetic side, analysis of the ratios β_j/β_i (105 ratios tested) shows membership in the quadratic field $\mathbb{Q}(\sqrt{2})$ to within 5×10^{-4} . It is possible to approximate the β_k in the compact form $\beta_k \approx (a_k + b_k\sqrt{2})/177$, with a maximum relative error of 4.9×10^{-5} . However, the LCM of the optimal free denominators reaches $\sim 10^{53}$, which demonstrates that 177 is not an exact geometric invariant, but a **pragmatic choice** minimizing the approximation error while preserving readability of the integer coefficients (a_k, b_k) . *The numerical values in Table III constitute the operational reference for all calculations presented in this paper.*

The model does not claim to **replace** quantum mechanics, quantum electrodynamics (QED), or quantum chromodynamics (QCD). It provides no information on wavefunctions, multipole moments, cross sections, or excitation spectra. Rather, it proposes a **numerical change of basis** that captures energy scales with high empirical precision, and whose structure suggests a deep link –still to be elucidated –with the geometry of exceptional lattices and the discrete symmetry of space.

As it stands, the β_k should be considered **universal phenomenological constants**, analogous in epistemological status to the Rydberg constant or the proton/electron mass ratio, but whose primary theoretical origin (an *ab initio* derivation from first principles) remains an open question. Their interdisciplinary universality and the statistical robustness validated by the randomization tests (section~10.4) invite further investigation towards an exact algebraic characterization and possible integration into a unified theoretical framework.

11.4 Limited scope of the model

Our model is designed solely for predicting masses and energies. It provides no information on wavefunctions, multipole moments, cross sections, or excitation spectra. Comparisons with established models (DFT, FRDM, Bethe-Weizsäcker) must be interpreted within this restricted scope : those models predict many observables, whereas ours handles only masses and energies.

11.5 Limitations and perspectives

- **Empirical nature** : no theoretical derivation of the β_k from first principles has yet been established.
- **Prediction of ε coefficients** : for a new system, ε must be extracted by fitting. A closed formula in terms of Z and A would be desirable.
- **Superheavy nuclei & new materials** : the model can predict unknown masses and gaps (see §5.7 and Appendix E).
- **Biology & AI** : the ternary code could be involved in molecular self-assembly and serve as a standard initialization method for deep networks.
- **Theory** : deriving the β_k from *zitterbewegung*, the Platonic solids, and E_8 algebra remains a major goal.

12. Conclusion

In this paper, we have discovered and validated a universal arithmetic quantization law that applies to atomic masses, ionization energies, covalent bonds, DNA base pairs, protein interactions, ATP energy, semiconductor band gaps, and the initialization of ternary neural networks. This law is written :

$$E = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \varepsilon_k \beta_k$$

with $\Lambda \in \{7.726 \text{ MeV}, 5.950 \text{ eV}\}$, $m \in \mathbb{Z}$, $\varepsilon_k \in \{-1, 0, 1\}$ and β_k the 15 universal constants from Table III. The precision obtained is exceptional ($< 0.001\%$ in all tested domains).

This result implies :

1. **Universality** : the same 15 numbers describe phenomena of very different natures.
2. **Cubic symmetry** : the binary octahedral group and the geometry of $\text{Cl}(6, 0)$ confirm an underlying discrete structure of space.
3. **Ternary code** : nature uses a 15-triplet code to “write” masses and energies.
4. **Practical applications** : prediction of exotic nuclei, discovery of materials, optimization of AI.
5. **Theoretical openings** : link between nuclear physics, lattice theory, combinatorics, and Lie algebras.

Nature, at all scales, obeys a discrete code whose rules we are only beginning to decipher.

13. Appendices

Appendix A –Decomposition of the 48 roots in the basis of the β_k

Appendix B –Use of AME2020 data for predicting isotopic masses

B.1. Files used

File	Description
mass.mas20.txt	Original AME2020 table (fixed Fortran format)
ame2020_VERIFIED.csv	Cleaned CSV version (generated by script)
isotope_fits_full.txt	Prediction results (output)
isotope_errors.png	Graph of relative errors

B.2. Data extraction (CSV reading)

The important columns are :

Index	Column	Content
2	Z	Atomic number
3	A	Mass number
12	AtomicMass_u	Atomic mass in micro-u (μu)

TABLE 17 – Decomposition of the 48 roots $\sqrt{\lambda_i}$ onto the 15 β_k constants

i	$\sqrt{\lambda_i}$	Error (%)	Active β_k coefficients	i	$\sqrt{\lambda_i}$	Error (%)	Active β_k coefficients
1	0.066136959	0.00×10^0	0	25	1.388242400	1.86×10^{-3}	+6 – 10 – 12 + 14
2	0.104861556	1.31×10^{-2}	–1 + 5 – 8 – 9 + 10	26	1.459652785	1.58×10^{-3}	–1 + 3 + 5 + 6 + 13 – 14
3	0.105819546	6.84×10^{-2}	+0 – 2 – 3 – 8 – 11 + 14	27	1.504114125	0.00×10^0	4
4	0.135301931	2.16×10^{-2}	+0 – 1 – 2 – 3 – 4 + 10	28	1.582953046	7.30×10^{-4}	+0 – 3 – 5 + 7 + 8
5	0.171977681	9.91×10^{-4}	+1 – 3 – 6 + 7 – 10 + 11	29	1.725183114	0.00×10^0	5
6	0.193286510	1.6×10^{-2}	–1 + 3 – 4 + 6 + 11 – 12	30	1.831271203	0.00×10^0	6
7	0.217318285	9.83×10^{-3}	–0 + 2 – 3 + 10 + 13 – 14	31	1.841477693	6.49×10^{-4}	+0 + 3 – 9 + 10 – 12 + 13
8	0.251706536	3.82×10^{-3}	–1 – 2 + 3 – 4 + 5	32	1.891359664	3.56×10^{-4}	–6 – 7 + 11 + 12 + 13 – 14
9	0.281751380	0.00×10^0	1	33	2.017424480	0.00×10^0	7
10	0.308944245	3.35×10^{-3}	–0 – 1 + 3 – 4 – 10 + 12	34	2.092834951	0.00×10^0	8
11	0.342536900	3.60×10^{-3}	–0 + 2 + 3 + 5 + 6 – 12	35	2.304598960	1.37×10^{-3}	+0 – 4 + 5 + 7
12	0.360145691	5.68×10^{-3}	–0 – 2 – 5 – 7 + 13	36	2.524926754	0.00×10^0	9
13	0.389044757	3.85×10^{-3}	–1 – 3 + 5 + 6 – 7	37	2.911315620	2.16×10^{-3}	+1 + 2 + 4 + 6 + 7 – 10
14	0.426899914	3.88×10^{-3}	+0 + 3 + 7 – 9	38	3.157658906	7.60×10^{-5}	–0 – 1 + 2 + 4 – 10 + 13
15	0.539423867	5.25×10^{-3}	+5 – 8 – 9 + 10 – 12 + 13	39	3.158280845	1.38×10^{-4}	–0 + 3 + 4 – 6 – 13 + 14
16	0.555869353	0.00×10^0	2	40	3.279177783	0.00×10^0	10
17	0.611492405	1.35×10^{-3}	–0 + 4 + 5 – 6 + 11 – 12	41	3.289042432	8.92×10^{-4}	+2 – 4 – 5 + 10 – 13 + 14
18	0.628435246	6.76×10^{-3}	+2 + 6 + 8 – 11	42	3.851497776	0.00×10^0	11
19	0.726794491	9.00×10^{-5}	–0 – 4 + 5 + 7 + 10 – 13	43	4.571886169	0.00×10^0	12
20	0.783638283	1.96×10^{-3}	+1 – 3 + 7 – 8 – 10 + 13	44	4.724739150	0.00×10^0	13
21	0.868248673	0.00×10^0	3	45	4.755055358	4.78×10^{-4}	–1 – 2 – 4 + 9 + 12
22	0.894790972	1.74×10^{-3}	–3 + 4 – 5 + 6 – 12 + 13	46	5.943089000	6.49×10^{-4}	–0 + 5 – 6 + 10 – 12 + 14
23	0.895000741	6.33×10^{-4}	+0 – 1 + 3 + 4 + 7 – 10	47	7.408061012	0.00×10^0	14
24	1.181361718	9.39×10^{-4}	+0 + 1 – 2 – 4 – 6 + 13	48	8.102307717	1.7×10^{-3}	–0 – 2 + 10 – 11 + 12 + 13

Conversion : the atomic mass in μu must be divided by (10^6) to obtain u , then multiplied by (931.49410242) MeV/ u .

B.3. Model parameters

Symbol	Value	Meaning
Λ	7.726 MeV	Fundamental nuclear scale constant
β_k	15 values	Universal constants (15 roots of Λ_{72})
Δ	$\sum \varepsilon_k \beta_k$	Ternary combinations ($\varepsilon \in \{-1, 0, 1\}$)
m	integer $\in [-5, 15]$	Scale exponent (factor 4^m)

B.4. Prediction algorithm

For each isotope (Z, A) of mass M_{true} :

1. Loop m from -5 to 15
2. Compute scale factor $f = 4^m$
3. Compute target $\tau = M_{\text{true}}/(\Lambda \cdot f)$
4. Find Δ closest in the list of ternary combinations
5. Compute $M_{\text{pred}} = \Lambda \cdot f \cdot \Delta$
6. Keep the pair (m, Δ) minimizing the relative error

B.5. Results obtained

Metric	Value
Isotopes processed	295
Mean error	0.0287 %
Standard deviation	0.0285 %
Maximum error	0.2458 %
% with error < 0.2%	100 %

B.6. Basic Python code

```
import numpy as np
from itertools import combinations, product

# Constants
Lambda = 7.726 # MeV
beta = np.array([...]) # 15 universal constants
```

```

# Generate ternary combinations  $\Delta = \varepsilon \cdot \beta$ 
def generate_delta_values(max_active=6):
    n = len(beta)
    deltas = []
    coeffs_list = []
    for k in range(1, max_active+1):
        for idx in combinations(range(n), k):
            for signs in product([-1, 1], repeat=k):
                coeff = np.zeros(n)
                val = 0.0
                for i, s in zip(idx, signs):
                    coeff[i] = s
                    val += s * beta[i]
                if val > 0:
                    first = next(i for i in range(n) if coeff[i] != 0)
                    if coeff[first] < 0:
                        coeff = -coeff
                        val = -val
                    deltas.append(val)
                    coeffs_list.append(coeff)
    order = np.argsort(deltas)
    return np.array(deltas)[order], [coeffs_list[i] for i in order]

```

```

delta_vals, coeffs_vals = generate_delta_values(max_active=6)

```

```

def predict_mass(M_true):
    best_err = float('inf')
    best_m = None
    best_pred = None
    best_delta = None
    best_coeff = None
    for m in range(-5, 16):
        factor = 4 ** m
        target = M_true / (Lambda * factor)
        if target < 0.01 or target > 20:
            continue
        idx = np.searchsorted(delta_vals, target)
        if idx == len(delta_vals):
            idx = len(delta_vals) - 1
        for candidate_idx in (idx, idx-1):
            if candidate_idx < 0 or candidate_idx >= len(delta_vals):
                continue
            delta = delta_vals[candidate_idx]
            pred = Lambda * factor * delta
            err = abs(pred - M_true) / M_true
            if err < best_err:
                best_err = err
                best_m = m

```

```

        best_pred = pred
        best_delta = delta
        best_coeff = coeffs_vals[candidate_idx]
    return best_m, best_delta, best_coeff, best_pred, best_err

```

B.7. Reading the AME2020 CSV

```

def read_isotopes(csv_file):
    isotopes = []
    with open(csv_file, 'r') as f:
        reader = csv.reader(f)
        next(reader) # skip header
        for row in reader:
            Z = int(row[2])
            A = int(row[3])
            # Mass in  $\mu u \rightarrow u \rightarrow \text{MeV}$ 
            mass_u = float(row[12].replace(' ', '').replace('#', '')) / 1e6
            mass_MeV = mass_u * 931.49410242
            isotopes.append((Z, A, mass_MeV))
    return isotopes

```

B.8. Important remarks

Symbol	Meaning
#	Estimated mass (not experimental)
*	Non-calculable value
Spaces in numbers	To be removed (e.g., 1 008664.91590 $\rightarrow$ 1008664.91590)
Neutron	Treated as (Z=0, A=1)

B.9. Generated outputs

File	Content
isotope_fits_full.txt	Complete table : Z, A, M_true, m, Δ , M_pred, err(%)
isotopes_epsilon.csv	Table with ϵ coefficients (15 columns)
isotope_errors.png	Error vs mass graph (log scale)

B.10. Physical interpretation

The exceptional precision (0.0287% on average, no deviation > 0.25%) suggests that nuclear masses follow a **discrete scaling law** :

$$M = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \varepsilon_k \beta_k$$

with $\Lambda = 7.726$ MeV, consistent with the binding energy per nucleon of medium nuclei ($\sim 7 - 8$ MeV). The 15 constants β_k are identical to 15 of the 48 non-degenerate roots of the exceptional lattice Λ_{72} , which establishes a deep link between the structure of nuclear masses and the geometry of exceptional lattices.

Appendix C –Coefficients ε_k and exponents m for ionization energies

The file **resultats_ionisations.csv** contains, for each pair (Z, k) , the exponent m and the 15 coefficients ε_k . Excerpt :

TABLE 24 – Ionization energies – Predictions and ε_k signatures

Z	k	Symbol	E_{exp} (eV)	E_{pred} (eV)	m	Δ	Err (%)	Signature ε_k
1	0	H	13.598434599702	13.598408443638	-1	9.141787189	0.000192	+0 – 3 + 8 + 10 + 12
1	0	H	13.602134636569	13.602100632838	-1	9.144269333	0.000250	+3 – 4 + 9 + 12 – 13 + 14
1	0	H	13.603365719000	13.603266861100	-1	9.145053352	0.000727	+0 – 1 + 9 + 10 – 11 + 14

The complete file (~6000 lines) is available as supplementary material.

Appendix D –Testable predictions

Masses of superheavy isotopes

TABLE 25 – Predictions for as-yet-unmeasured superheavy isotopes

Z	A	Element	M_{pred} (MeV)	Signature ε
110	279	Ds	259742.3	–0 – 1 + 2 + 4 – 7 + 9 + 11 – 13
110	280	Ds	260151.7	+0 + 3 – 5 + 8 + 10 – 12 – 14
111	282	Rg	262584.1	–1 + 2 – 4 + 6 – 8 + 11 + 13

First ionization energies of superheavy elements

Appendix E –Python scripts

The scripts used are available online (GitHub/Zenodo repository) : - `extract_ame2020.py` : extraction and conversion of masses - `generate_epsilon.py` : generation of ternary combinations - `optimize_beta.py` : optimization of β_k and Λ - `ternary_nn.py` : training of the ternary network - `plot_accuracy_loss.py` : performance visualization

TABLE 26 – Predictions for first ionization energies ($Z = 105$ to 110)

Z	Element	E_{pred} (eV)	Signature ε
105	Db (Dubnium)	8.12	+0 + 2 − 5 + 7 + 9 − 11 + 14
106	Sg (Seaborgium)	8.45	+1 − 3 + 6 + 8 − 10 + 12
107	Bh (Bohrium)	8.78	−0 + 2 − 4 + 7 + 9 − 13 + 14
108	Hs (Hassium)	9.02	+1 + 3 − 5 − 8 + 11 − 13
109	Mt (Meitnerium)	9.31	−0 − 2 + 4 + 6 − 9 + 12 + 14
110	Ds (Darmstadtium)	9.55	+0 + 1 − 3 − 5 + 8 + 10 − 12

14. References

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